Allamaprabhu Ani

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scholar.google.com/citations?user=dX3xKGsAAAAJ

Research Statement

PhD student developing GPU native, autograd-driven phase field fracture solvers with neural operator constitutive laws. Research bridges computational mechanics and machine learning: differentiable physics for inverse problems, learned closures for multiscale fracture, and benchmark methodology for the field. Lead author of the 2026 *Engineering Fracture Mechanics* review (Vol. 332, Art. 111778) on machine learning for fracture mechanics with collaborators at EPFL, University of Florida, and City St George's.

Education

PhD, Mechanical Engineering Jul 2023 – Jun 2026 (expected)
City, St George's, University of London – School of Sciences and Technology
Thesis: *Machine-learning-enabled computational fracture and damage mechanics: differentiable GPU solvers, inverse problems, and learned constitutive models*.
Supervisor: Dr Sathiskumar A. Ponnusami FRAeS.

MTech, Design and Manufacturing Sep 2021 – May 2023
National Institute of Technology Silchar – CGPA 9.88/10. Best Student Award.
Thesis: Machine learning for fatigue life prediction in aluminium alloys. Advisor: Dr A.B. Deoghare.

BE, Mechanical Engineering Aug 2016 – Jul 2020
GM Institute of Technology, Davangere (Visvesvaraya Technological University) – CGPA 8.47/10.

Awards and Honours

- **Yeoman and Travelling Scholarship, Worshipful Company of Tin Plate Workers** (Mar 2026). One of three Travelling Scholarship recipients across the City of London, for "AI-accelerated modelling for fracture prediction."
- **Dean's Award for Outstanding Teaching Support**, School of Sciences and Technology, City St George's, University of London (Apr 2026). Nominated by students.
- **Associate Fellow of the Higher Education Academy (AFHEA)**, Advance HE (Oct 2024). Credential PR301728.
- **Funded PhD Studentship** "Modelling for Failure Analysis," City St George's, University of London (2023).
- **Best Student Award**, MTech Design and Manufacturing, National Institute of Technology Silchar (2023).
- **Inspiring Student Award for Mechanical Engineering**, Visvesvaraya Technological University (Dec 2020).

Publications

Journal articles

J1. **Ani, A.S.**, Nakka, R., Subhash, G., Molinari, J.-F., Ponnusami, S.A. (2026). Machine learning for computational fracture and damage mechanics: Status and perspectives. *Engineering Fracture Mechanics*, 332, 111778. doi:10.1016/j.engfracmech.2025.111778. *Open access; field-level review paper.*

J2. Srinivasa, C.V., **Ani, A.S.** (2020). Protective Coatings for Bio-Composites: A Review. *IOP Conference Series: Materials Science and Engineering*, 925, 012048. doi:10.1088/1757-899X/925/1/012048.

Book chapters and conference papers

C1. **Ani, A.S.**, Deoghare, A.B. (2024). Leveraging Machine Learning for Enhanced Fatigue Life Prediction in Aluminium Alloys. In: *Lecture Notes in Mechanical Engineering*, Springer. doi:10.1007/978-981-97-7535-4_61.

In preparation

- Two-part submission to *Computer Methods in Applied Mechanics and Engineering* (CMAME) on differentiable phase field fracture and autograd-based inverse problems. Drafts in revision; submission target Q3 2026.

Research Experience

Doctoral Researcher, Computational Mechanics and Machine Learning Jul 2023 – present
City, St George's, University of London & Queen Mary University of London

- Designed and built `torch_pf_solver`: a PyTorch native, GPU-accelerated, autograd-differentiable phase field fracture solver with neural operator hooks for learned constitutive laws.
- Demonstrated inverse problems via gradient based optimisation: scalar fracture toughness recovery, spatial Gc-field reconstruction, and stiff inclusion localisation, all driven by autograd through a staggered solver.
- Cross-validated against Akantu (C++), FEniCSx, COMSOL, PhaseFieldX and PhaFiDyn on quasi static and dynamic benchmarks (Borden, Bourdin, Miehe, Kalthoff, SENT, Brazilian disc).
- Maintained reproducible HPC workflow on Hyperion2 (Slurm) with local Mac CPU and WSL CUDA development environments.

Visiting Researcher, LSMS Lab, EPFL 2025
Host: *Prof. Jean-François Molinari* (École Polytechnique Fédérale de Lausanne).

- Collaborated on the *Engineering Fracture Mechanics* 2026 review of machine learning for computational fracture and damage mechanics.
- Engaged with the LSMS group on differentiable methods for solid mechanics; visit informed the autograd-through-staggered-solver framework in the PhD code base.

MTech Researcher, National Institute of Technology Silchar 2021 – 2023
Advisor: *Dr A.B. Deoghare*.

- Machine-learning surrogate models for fatigue life prediction across multiple aluminium alloy compositions (scikit-learn, MATLAB). Best Student Award.

Founder and Tool Developer, Aeroknacks India Ltd 2020 – 2021

- Built a library of validated aerospace hand-calculation automation tools and Excel-VBA workflows based on E.F. Bruhn, Michael Niu, Boeing Design Manuals, and ESDU references.
- Implemented BJSFM (Lekhnitskii anisotropic stress field with de Jong correction), Cozzone f_0 plastic-bending extraction, lug strength (Bruhn/Niu/ESDU), fastener lap-joint load transfer, column buckling (Euler/Johnson + AFFDL coefficients), modal analysis (Roark validation), ABD calculator, Neuber non-linear stress, and a NASTRAN `.f06` parser. Tools shipped to an aerospace engineering firm.

Research Intern, NMCAD Lab, Indian Institute of Science Bengaluru
Research Intern, CGPL Lab, Indian Institute of Science Bengaluru (Feb 2018) – Pressure Swing Adsorption for 99.997%-purity hydrogen from biomass gasifier.

Teaching and Outreach

Teaching, City St George's (2023 – present)

- Co-pilot, Learning Enhancement and Development (LEaD) educational technology team (Sep 2024 – Feb 2025).
- Graduate Teaching Assistant, EG1004 Introduction to Mechanics, Materials and Manufacturing.
- Graduate Teaching Assistant, EG1002 Wind Turbine Challenge: Tower Design, and EG1002 Siemens NX CAD teaching support.
- Head Invigilator, School of Sciences and Technology (Dec 2023 – Apr 2024); Regular Invigilator, Examinations Office (Apr 2023 – Jan 2025).
- "Engineering 2050 for Good" Hackathon (Oct 2023).

Teaching, National Institute of Technology Silchar (2021 – 2023)

- Teaching Assistant: ME-316 Machine Design Laboratory (SolidWorks, ANSYS, FEA); ME-202 Theory of Machines.

Workshop co-organisation

- Co-organiser, M2L (Mechanics and Machine Learning) Workshop, City St George's, July 2024 and July 2025. EPSRC-supported. Speakers: M. Khodaei (Imperial), G. Subhash (Florida), J.-F. Molinari (EPFL), S. Kumar

(TU Delft), S. Chakraborty (IIT Delhi), S.A. Ponnusami (City).

Open-Source Software and Online Presence

- **torch_pf_solver** – PyTorch native differentiable phase field fracture solver (research code; selective release planned alongside publications).
- **bjsfm** – github.com/allamaprabhuani/bjsfm – anisotropic stress field around a hole in a composite plate (Lekhnitskii, de Jong). Public release with attribution to upstream MIT-licensed source.
- **geometric-transformations** – github.com/allamaprabhuani/geometric-transformations – 2D/3D affine transforms with MATLAB Live Scripts.
- **Personal site** – allamaprabhuani.github.io – research notes, technical tutorials, blog.

Technical Skills

Scientific computing and ML. PyTorch (autograd, CUDA), JAX (familiar), NumPy/SciPy, scikit-learn, pandas. Differentiable physics, neural operators (DeepONet, FNO), physics-informed neural networks.
Finite element analysis. Akantu (C++), FEniCSx, COMSOL Multiphysics, MSC Nastran/Patran (SOL-101/103/105/106), Altair Hypermesh/Optistruct, Abaqus, ANSYS, SolidWorks.
HPC and infrastructure. Slurm, Linux clusters, Git, Sphinx, latexmk, reproducible Python packaging (pyproject, editable installs).
Programming. Python, MATLAB, C++ (read), Excel-VBA, LaTeX, Bash.
Reference texts and manuals. E.F. Bruhn (Flight Vehicle Structures), Michael Niu (Airframe Stress Analysis), Boeing Design Manuals, ESDU, Timoshenko (Plates and Shells), Lekhnitskii (Anisotropic Plates), Roark (Stress and Strain).

Referees

Available on request.